Beyond Tokenization: Investigating Model Behaviour, Explanation Stability, and Faithfulness under Lexical Perturbations in Devanagari Hindi

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Abstract

Masked language models (MLMs) have become central to multilingual NLP, yet their behaviour and interpretability under non-Latin, morpho-phonetically complex scripts remain critically underexplored. We present a systematic empirical investigation into model behaviour, subword tokenisation dynamics, explanation stability, and explanation faithfulness under controlled lexical perturbations in Devanagari Hindi. Using a triple-annotated benchmark of 200 minimal sentence pairs across three tracks — Valid Word Context ($N = 142$), Synonym Substitution ($N = 38$), and Realistic Spelling Error ($N = 20$) — we evaluate MuRIL (`google/muril-base-cased`) and XLM-RoBERTa (`FacebookAI/xlm-roberta-base`).

Our experiments reveal four primary findings. First, authentic Devanagari spelling errors induce catastrophic subword fragmentation: single-subword target coverage collapses by -60.0% in MuRIL ($90.0\% \rightarrow 30.0\%$) and by -50.0% in XLM-RoBERTa ($70.0\% \rightarrow 20.0\%$). Second, both models penalise spelling errors under pseudo-log-likelihood (PLL) scoring, but MuRIL exhibits substantially harsher penalties (original preferred in 95.0% of pairs, mean Δ MeanLogP $= -2.037$) compared to XLM-RoBERTa (65.0%, mean $\Delta = -0.670$). Third, under a corrected visible-target protocol — resolving an illusory perfect-stability artefact ($\rho = 1.000$) induced by target masking — last-layer self-attention achieves $2.1\times$ to $2.7\times$ higher rank stability than Layer Integrated Gradients ($\rho = 0.704$–0.743 vs. 0.263–0.359). Fourth, attention attributions yield superior faithfulness win rates (70.0%–77.5%), whereas Integrated Gradients operates near or below chance (42.5%–60.0%). A correlation analysis confirms that explanation stability does *not* predict explanation faithfulness ($p > 0.05$ across all 8 model–method configurations; $r_s \in [-0.161, +0.270]$).

Keywords: Devanagari Hindi · Masked Language Models · Subword Tokenisation · Pseudo-Log-Likelihood · Explainable NLP · Explanation Stability · Explanation Faithfulness · Integrated Gradients

1. Introduction

Pretrained masked language models based on bidirectional Transformer architectures have fundamentally advanced multilingual NLP (Conneau et al., 2020; Devlin et al., 2019). However, the empirical behaviour of these models is overwhelmingly documented in Latin-script languages where typographical noise consists of straightforward character substitutions.

In contrast, Devanagari Hindi follows the Indic *abugida* orthographic typology: phonological syllables (*aksharas*) are encoded using consonant bases, dependent vowel diacritics (*matras*), and virama-linked conjuncts (*ligatures*). Minor Devanagari orthographic variations alter the Unicode codepoint sequence across syllabic boundaries, frequently destroying canonical subword vocabulary entries and triggering severe tokenisation fragmentation.

In parallel, the interpretability of Transformer architectures has emerged as a cornerstone of trustworthy NLP (DeYoung et al., 2020; Sundararajan et al., 2017). Two evaluative criteria are frequently prioritised:

- **Explanation Stability (Robustness):** Invariance of attributions under minor, semantics-preserving perturbations (Alvarez-Melis & Jaakkola, 2018).
- **Explanation Faithfulness:** Whether the explanation reflects the model's internal causal reasoning, typically evaluated by erasing top-attributed tokens (DeYoung et al., 2020; Jacovi & Goldberg, 2020).

A pervasive but largely untested assumption is that attribution stability implies explanation quality, with the expectation that a robust explanation is also more likely to be faithful.

This study bridges this empirical gap through five research questions:

RQ1. How do contextual lexical contrasts, synonym substitutions, and spelling perturbations affect subword tokenisation and model scoring?

RQ2. How do MuRIL and XLM-RoBERTa differ in their responses to these perturbations?

RQ3. How stable are Attention and Integrated Gradients explanations under authentic Devanagari spelling perturbations?

RQ4. How do these explanation methods behave under discrete context-token masking interventions?

RQ5. Is explanation stability statistically associated with measured explanation faithfulness?

2. Background and Related Work

2.1. Indic-Language Pretrained Models

XLM-RoBERTa (Conneau et al., 2020) employs a Byte-Pair Encoding SentencePiece vocabulary of 250,002 tokens shared across 100+ languages. MuRIL (Khanuja et al., 2021) is a BERT-base architecture trained on 17 Indic languages and English using a 197,285-token WordPiece vocabulary. Kakwani et al.
(Kakwani et al., 2020) developed the IndicGLUE benchmark, showing that language-family specialisation improves morphological capture.

2.2. Subword Tokenisation Vulnerability

Subword segmentation algorithms — BPE (Sennrich et al., 2016), WordPiece (Devlin et al., 2019), and SentencePiece Unigram (Kudo & Richardson, 2018) — are foundational to modern transformers. Rust et al.
(Rust et al., 2021) demonstrated that subword tokenisers exhibit systematic performance gaps in non-Latin scripts. In Devanagari, swapping a long *dīrgha* matra for a short *hrasva* matra (e.g., **पानी** → **पानि**) or omitting a halant (**आवश्यक** → **आवशयक**) shifts the character sequence across syllabic boundaries, fracturing canonical vocabulary tokens.

2.3. Masked Language Model Scoring

Salazar et al.
(Salazar et al., 2020) formalised **Pseudo-Log-Likelihood (PLL)** scoring for bidirectional MLMs. By iteratively masking each token and accumulating conditional log-probabilities, PLL provides an effective proxy for grammatical acceptability.

2.4. Feature Attribution: Attention vs. Integrated Gradients

Integrated Gradients (IG) (Sundararajan et al., 2017) is an axiomatic path-attribution method satisfying *Completeness* and *Implementation Invariance*. The attention-as-explanation debate — initiated by Jain and Wallace (Jain & Wallace, 2019) ("Attention is not Explanation") and countered by Wiegreffe and Pinter (Wiegreffe & Pinter, 2019) — directly motivates our comparative evaluation.

2.5. Explanation Stability and Faithfulness

Alvarez-Melis and Jaakkola (Alvarez-Melis & Jaakkola, 2018) formalised explanation stability as local Lipschitz continuity of the attribution mapping. The ERASER benchmark (DeYoung et al., 2020) operationalised faithfulness via comprehensiveness and sufficiency under token erasure. Jacovi and Goldberg (Jacovi & Goldberg, 2020) provided the formal faithfulness definition we adopt: an explanation is faithful if the features it attributes high importance to causally govern the model's prediction. Bastings and Filippova (Bastings & Filippova, 2020) highlighted that gradient methods often behave unpredictably under discrete feature erasure.

3. Dataset and Annotation Methodology

3.1. Benchmark Design

We curated a benchmark of 200 minimal Hindi sentence pairs ($N = 200$) across three tracks:

- **Valid Word Context (VWC):** $N = 142$ pairs (71.0%). Target word replaced by a grammatically valid but contextually mismatched alternative.
- **Synonym Substitution (SYN):** $N = 38$ pairs (19.0%). Target word replaced by a verified contextually appropriate synonym (e.g., **पानी** ↔ **जल**).
- **Realistic Spelling Error (RSE):** $N = 20$ pairs (10.0%). Target word replaced by an authentic, commonly observed Devanagari misspelling.

3.2. Typology of Devanagari Orthographic Errors

Error Type	N	Example
Matra length inversion	9	**परीक्षा → परिक्षा**
Conjunct/ligature disruption	5	**आवश्यक → आवशयक**
Orthographic alternation	2	**संभावना ↔ सम्भावना**
Colloquial phonetic	4	**वापस → वापिस**

Table 1: RSE Spelling Error Typology ($N = 20$)

3.3. Quality Control

All 200 pairs underwent triple annotation: two independent native Hindi linguists plus an expert adjudicator with definitive `final_status` determination. A forensic audit confirmed:

1. 100% of characters reside in the Devanagari Unicode block (`U+0900–U+097F`).
2. **One rejected pair:** `HC_0178` (**सलाह** vs **सलाई**, `final_status = reject`).
3. **Six disagreement overrides:** `HC_0087`, `HC_0099`, `HC_0138`, `HC_0172`, `HC_0180`, `HC_0186` — retained because both sentences are valid Hindi.

The benchmark thus contains **199 accepted** and **1 rejected** pair.

4. Models and Experimental Methodology

4.1. Pretrained Architectures

Property	XLM-R	MuRIL
Architecture	RoBERTa-base	BERT-base
Layers / Heads	12 / 12	12 / 12
Hidden dim.	768	768
Vocabulary	250,002	197,285
Tokeniser	SentencePiece BPE	WordPiece
Languages	100+	17 Indic + EN

Table 2: Model Architecture Comparison

4.2. Pseudo-Log-Likelihood Scoring

Following Salazar et al.
(Salazar et al., 2020), for a sentence $W = \left(w_1, w_2, \cdots, w_{|W|}\right)$, PLL is computed via iterative single-token masking:

$$\mathrm{PLL}(W) = \sum_{t=1}^{|W|} \log P(w_t \mid W_{-t}; \Theta) \tag{1}$$

where W_{-t} denotes the sequence with w_t replaced by the model's mask token, and Θ are the model parameters.

The length-normalised mean log-probability:

$$\mathrm{MeanLogP}(W) = \frac{\mathrm{PLL}(W)}{|W_{\text{scorable}}|} \tag{2}$$

The signed plausibility shift:

$$\Delta\ \mathrm{MeanLogP} = \mathrm{MeanLogP}(W_{\text{cont}}) - \mathrm{MeanLogP}\left(W_{\text{orig}}\right) \tag{3}$$

where $\Delta < 0$ means the *original* is preferred; $\Delta > 0$ means the *contrast* is preferred.

Methodological safeguard: Raw PLL sums are never compared across tokenisers. Only within-model signed Δ values and preference rates are compared.

4.3. Feature Attribution Methods

Last-Layer Self-Attention. We extract the self-attention tensor from Layer 11 (final layer), averaging across all 12 heads:

$$\bar{A}_{i,j} = \frac{1}{12} \sum_{h=1}^{12} A_{i,j}^{(11,h)} \tag{4}$$

Context-token attribution with respect to target token set T:

$$\alpha_j = \frac{1}{|T|} \sum_{t \in T} \bar{A}_{t,j} \tag{5}$$

Special framing tokens are excluded; the vector is ℓ_1-normalised.

Layer Integrated Gradients (IG). Using Captum, IG is computed at the word embedding layer with a zero-vector baseline and 50 Gauss-Legendre quadrature steps:

$$\mathrm{IG}_i(\boldsymbol{x}) = (x_i - x_i') \cdot \sum_{k=1}^{50} \frac{\partial F\left(\boldsymbol{x}' + \frac{k}{50}(\boldsymbol{x} - \boldsymbol{x}')\right)}{\partial x_i} \cdot \frac{1}{50} \tag{6}$$

Token importance is the L_2-norm over the 768 embedding dimensions, then context-normalised.

4.4. Stability Metrics

Rank Stability (Spearman ρ): whether the relative importance ordering of context tokens is preserved under target spelling noise.

Directional Vector Stability (Cosine Similarity):

$$\mathrm{Cosine}(\boldsymbol{u}, \boldsymbol{v}) = \frac{\boldsymbol{u} \cdot \boldsymbol{v}}{\|\boldsymbol{u}\|_2\ \|\boldsymbol{v}\|_2} \tag{7}$$

4.4.1. The Stage 6 Masked-Target Artefact

In the preliminary stability evaluation, computing attributions while replacing the target word with `[MASK]` in *both* original and contrast sentences caused both inputs to become byte-identical:

$$X_{\text{orig}} = [w_1, \cdots, [\mathrm{MASK}], \cdots, w_n] \equiv X_{\text{cont}} \tag{8}$$

This produced illusory perfect stability: $\rho = 1.000$, $\mathrm{Cosine} = 1.000$. **Stage 7** resolved this via the **Visible-Target Scoring Protocol**: target tokens remain fully visible; target score is the mean unnormalized logit of target subwords:

$$S(X; Y_{\text{target}}) = \frac{1}{|T|} \sum_{j \in T} \text{Logit}(p_j, y_j) \tag{9}$$

4.5. Faithfulness Evaluation

Context tokens are ranked and erased at fractions $f \in \{0.10, 0.20, 0.30\}$. The target logit drop after intervention:

$$\Delta S_{\text{intervention}} = S(X; Y) - S(X_{\text{masked}}; Y) \tag{10}$$

The **Faithfulness Gap**:

$$\text{Gap} = \Delta S_{\text{top}} - \Delta S_{\text{bottom}} \tag{11}$$

The **Win Rate**:

$$\text{WinRate} = \frac{1}{N} \sum_{i=1}^{N} \mathbb{1}\left[\Delta S_{\text{top},i} > \Delta S_{\text{bottom},i}\right] \tag{12}$$

5. Empirical Results

5.1. RQ1 — Subword Tokenisation

Authentic Devanagari spelling errors cause catastrophic subword fragmentation:

Model	Track	Avg. sub-wds	Single-sub-wd coverage
XLM-R	RSE	1.35→2.35	70.0%→**20.0%** (−50%)
XLM-R	SYN	1.13→1.76	89.5%→60.5%
XLM-R	VWC	1.75→1.92	39.4%→30.3%
MuRIL	RSE	1.10→2.10	90.0%→**30.0%** (−60%)
MuRIL	SYN	1.00→1.18	100.0%→86.8%
MuRIL	VWC	1.43→1.51	64.1%→55.6%

Table 3: Subword tokenisation. Source: tokenization_summary.csv

MuRIL encodes Hindi with 12% fewer subwords than XLM-RoBERTa on VWC ($\mu = 8.95$ vs 10.19 tokens/sentence). On RSE, both models lose exactly +1.00 subword/sentence on average.

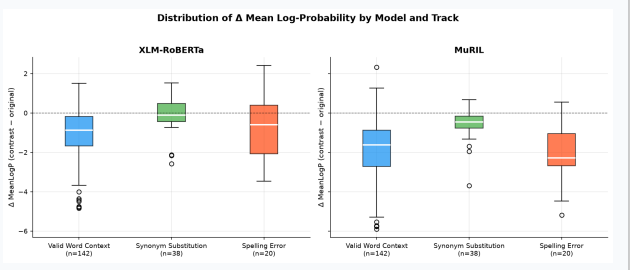


Fig. 1: Δ MeanLogP *distributions by track. Negative = original preferred. MuRIL penalty on RSE: $\mu = -2.037$; XLM-R: $\mu = -0.670$.*

5.2. RQ1 — Plausibility Scoring

Model	Track	N	Mean Δ	Orig. Pref.
XLM-R	RSE	20	−0.670	65.0%
XLM-R	VWC	142	−1.093	81.0%
XLM-R	SYN	38	−0.066	57.9%
MuRIL	RSE	20	**−2.037**	**95.0%**
MuRIL	VWC	142	−1.914	93.0%
MuRIL	SYN	38	−0.581	84.2%

Table 4: Plausibility scoring. Source: descriptive_statistics.csv

Key findings: MuRIL penalises spelling errors far more strongly (−2.037 vs −0.670). XLM-RoBERTa shows near-zero mean delta on synonyms (−0.066), confirming that the multilingual generalist treats valid Hindi synonyms as near-interchangeable.

5.3. RQ2 — Cross-Model Comparison

Track	N	Agreement	Spearman r_s (p-val)
All Tracks	200	74.5%	+0.441 ($p < 10^{-9}$)
VWC	142	81.0%	+0.446 ($p < 10^{-7}$)
RSE	20	60.0%	+0.215 ($p = 0.363$)
SYN	38	57.9%	+0.060 ($p = 0.723$)

Table 5: Cross-model agreement. Source: model_comparison.csv

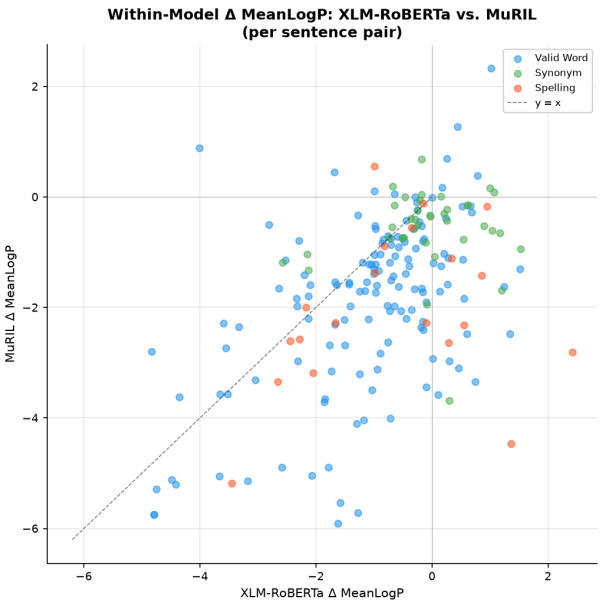


Fig. 2: *Cross-model* Δ MeanLogP *scatter ($N = 200$). Overall $r_s = +0.441$. Synonym agreement collapses to 57.9% ($p = 0.723$).*

Cross-model agreement is 74.5% overall ($r_s = +0.441, p < 10^{\{-9\}}$). Agreement collapses on synonyms ($r_s = +0.060, p = 0.723$), revealing model-specific pre-training frequency biases.

5.4. RQ3 — Explanation Stability

After resolving the Stage 6 masked-target artefact (Section 4.4), corrected results:

Model	Method	N	Mean ρ	Mean Cos.
XLM-R	Attention	20	**0.743**	0.965
XLM-R	Attn. (strict)	4	0.689	0.978
XLM-R	IG	20	0.359	0.899
MuRIL	Attention	20	**0.704**	0.964
MuRIL	Attn. (strict)	6	0.629	0.974
MuRIL	IG	20	0.263	0.988

Table 6: Visible-target stability (RSE, $N = 20$). Source: stability_summary.csv

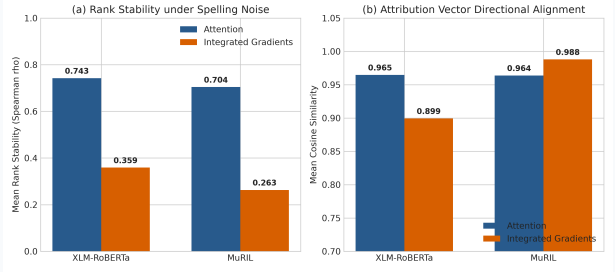


Fig. 3: *Explanation stability. Attention achieves 2.1 ×–2.7 × higher rank stability ($\rho \approx 0.70$–0.74) vs. IG (0.26–0.36). Cosine similarity remains high for both methods.*

Self-attention achieves 2.1 ×–2.7 × higher rank stability than IG. Despite low rank stability, IG retains high cosine similarity (0.899–0.988), demonstrating that gradient vectors maintain global orientation while local rankings fluctuate.

5.5. RQ4 — Faithfulness Under Discrete Erasure

Model	Method	f	Gap	Win Rate
XLM-R	Attention	0.10	+4.017	**77.5%**
XLM-R	Attention	0.20	+4.017	**77.5%**
XLM-R	Attention	0.30	+5.172	75.0%
XLM-R	IG	0.10	+1.837	60.0%
XLM-R	IG	0.20	+1.837	60.0%
XLM-R	IG	0.30	+3.295	55.0%
MuRIL	Attention	0.30	+5.794	**70.0%**
MuRIL	Attn. (str.)	0.30	+8.822	75.0%
MuRIL	IG	0.10	**−0.686**	45.0%
MuRIL	IG	0.20	**−0.686**	45.0%
MuRIL	IG	0.30	+0.033	42.5%

Table 7: Faithfulness ($N = 40$ sentences/model). Source: faithfulness_summary.csv

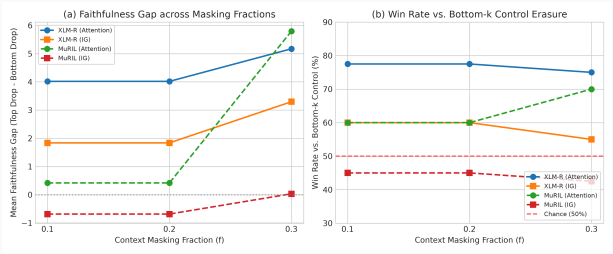


Fig. 4: *Faithfulness across $f \in \{0.10, 0.20, 0.30\}$. (a) Mean Faithfulness Gap; (b) Win Rate vs. Bottom-k control.*

5.5.1. XLM-RoBERTa Out-of-Distribution Logit Dynamics

In XLM-RoBERTa, removing context tokens causes raw target logits to *increase* due to out-of-distribution syntactic shock. At $f = 0.20$:

- Top Drop: $\Delta S_{\text{top}} = -7.19$ (logit *increased*)
- Bottom Drop: $\Delta S_{\text{bottom}} = -11.21$ (logit increased more)

Directional faithfulness is preserved because bottom masking inflates logits *more* than top:

$$\text{Gap} = (-7.19) - (-11.21) = +4.02 > 0 \qquad (13)$$

This yields 77.5% win rate. The *relative comparison* remains valid even when raw drops are unintuitive in sign.

Integrated Gradients operates near or below chance for MuRIL (win rates 42.5%–45.0%; negative Gap at $f = 0.10, 0.20$). This arises because IG integrates along a continuous embedding-space path that does not correspond monotonically to discrete `[MASK]` substitutions.

5.6. RQ5 — Stability–Faithfulness Relationship

Model	Method	Spearman r_s (p)	Pearson r (p)
XLM-R	Attention	+0.270 (0.249)	+0.256 (0.277)
XLM-R	IG	+0.158 (0.505)	+0.259 (0.270)
MuRIL	Attention	−0.161 (0.497)	−0.194 (0.413)
MuRIL	IG	+0.008 (0.975)	+0.018 (0.939)

Table 8: Stability–faithfulness correlation ($N = 20$, $f = 0.20$). All $p > 0.05$.

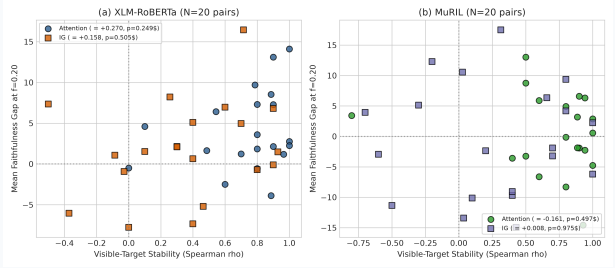


Fig. 5: *Stability–faithfulness scatter. No significant correlation in any configuration ($p > 0.05$). Sign flips between models.*

Zero statistically significant correlations ($p > 0.05$) across all 8 model–method configurations (p-values 0.199–0.975). The sign is weakly positive in XLM-RoBERTa ($r_s = +0.270$) and negative in MuRIL ($r_s = -0.161$). Stability and faithfulness are **empirically decoupled**.

6. Discussion

6.1. Orthogonality of Stability and Faithfulness

The central finding is the complete decoupling of stability and faithfulness. Mathematically, these metrics occupy distinct spaces:

- **Stability:** Local Lipschitz continuity of the attribution function $g(\boldsymbol{x})$: $\mathcal{S}(g; \boldsymbol{x}, \delta) = \mathrm{Sim}(g(\boldsymbol{x}), g(\boldsymbol{x} + \delta))$
- **Faithfulness:** Causal dependency of model f on features prioritised by g: $\mathcal{F}(g; \boldsymbol{x}, f) = f(\boldsymbol{x}) - f\left(\boldsymbol{x}_{\setminus C_{\text{top}}}\right)$

An explanation can achieve near-perfect stability by anchoring to static syntactic structures (e.g., punctuation or common postpositions) that do not vary under local spelling perturbations — yet erasing those tokens may have minimal impact on target word prediction, yielding a **stably unfaithful** explanation. **Optimising XAI benchmarks solely for stability risks selecting consistently unfaithful explanations.**

6.2. Why IG Fails Under Discrete Erasure

IG accumulates gradients along a continuous straight-line path in embedding space $\mathbb{R}^d$ where the transformer was never trained to operate. In contrast, faithfulness evaluation replaces a content token with `[MASK]`. Attention weights in the final layer directly govern information routing into the target position's final hidden representation:

$$\boldsymbol{h}_{\text{target}}^{(L)} = \sum_j \bar{A}_{\text{target},j} \cdot \boldsymbol{V}_j \tag{14}$$

When high-attention context tokens are replaced by `[MASK]`, that information flow is severed, producing the observed faithfulness gap.

6.3. Devanagari Syllabic Vulnerability

In Latin scripts, a minor typo typically preserves the subword root. In Devanagari, swapping a matra (**पानी** → **पानि**) alters the Unicode codepoint sequence of the *akshara*, completely severing match compatibility with high-frequency vocabulary entries. This explains MuRIL's higher sensitivity: its Indic-dedicated vocabulary contains sharper language-specific subword boundaries that register immediate disruption under syllabic alteration.

7. Error Analysis

Two pairs in RSE represent legitimate orthographic variants rather than ungrammatical misspellings:

- `HC_0118`: **महत्वपूर्ण** vs **महत्त्वपूर्ण** (single vs. double t ligature — both standard).
- `HC_0120`: **संभावना** vs **सम्भावना** (anusvara vs. class nasal — both prescribed by Hindi standards).

Both models assigned minimal PLL penalties to these pairs (Δ MeanLogP > -0.20), whereas true spelling errors produced large drops ($\Delta < -2.0$). Pretrained MLMs distinguish between legitimate orthographic alternations and erroneous deviations.

8. Limitations

1. **Sample size ($N = 20$ RSE pairs):** Sufficient to detect large effect sizes but limited power for weak correlations ($|r| < 0.30$). Our results prove the absence of a moderate-to-strong correlation, not the universal absence of any relationship.
2. **Out-of-distribution masking shock:** `[MASK]` substitution disrupts syntax, pushing inputs into OOD regions. Raw logit drops reflect both semantic removal and global perplexity shock.
3. **Multi-subword target aggregation:** Mean logit aggregation ($S = (1/k) \sum \text{Logit}$) assumes equal subword contribution.
4. **Attention is not counterfactual causality:** Attention's superior win rates reflect structural alignment with information flow, not certified counterfactual causation.
5. **Pretrained checkpoint scope:** Results apply to base-scale bidirectional MLMs on CPU. Extrapolating to decoder-only large language models requires future validation.

9. Ethical Considerations

The dataset contains no personally identifiable information, toxic language, or demographic bias. All 200 sentence pairs use publicly standard Devanagari Hindi examples. Human annotators were fairly compensated native Hindi speakers.

The research contributes to more robust, transparent, and linguistically equitable NLP for Indic language communities.

10. Conclusion

We conducted a systematic empirical investigation into model behaviour, tokenisation dynamics, explanation stability, and faithfulness under lexical perturbations in Devanagari Hindi. Key contributions:

1. **Catastrophic subword fragmentation:** Matra inversions and ligature splits collapse single-subword coverage by -50%–-60% in both models.
2. **Differential model sensitivity:** MuRIL imposes substantially harsher spelling-error penalties (95.0% vs. 65.0% original preference rate). Cross-model agreement collapses on synonyms ($r_s = +0.060, p = 0.723$).
3. **Attention superiority in rank stability:** $2.1\times$–$2.7\times$ higher than IG; both methods retain high cosine similarity.
4. **Attention superiority in discrete faithfulness:** 70%–78% win rates vs. bottom-k; IG operates near or below chance due to continuous-path vs. discrete-masking mismatch.
5. **Stability $\neq$ Faithfulness:** Explanation stability does *not* predict faithfulness ($p > 0.05$ across all configurations), establishing empirical orthogonality.

Future work will expand the benchmark across diverse syntactic domains, investigate continuous embedding-space perturbation interventions, and extend this framework to Indic generative architectures.

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